



SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY

An Autonomous Institution| Approved by AICTE| Affiliated to Anna University

Kuniamuthur, Coimbatore – 641008

Phone: (0422)-2628001 (7 Lines) | Email : info@skcet.ac.in | Website: www.skcet.ac.in

TOTAL QUALITY MANAGEMENT

Module - 1

1.1.Quality, Dimensions of Quality

What is quality?



Quantified definition of Quality:

$$Q = P / E$$

where,

Q = quality

P = performance

E = expectations



If Q is greater than 1, then the customer has a good feeling about the quality of product or service.

What is TQM?

TQM implies that ***every one*** associated with the organization is committed towards ***continual improvement*** of the organization through ***customer satisfaction***.



TQM - A Road to success

Dimensions of quality

Dimensions of quality are the different features of product or service.

Functionality:

Functionality refers to the core features and characteristics of a product that satisfy the customer.

Reliability:

Reliability is measured by mean time between failures (MTBF) and mean time to first failure. Reliability is an indicator of durability of products.

Usability:

A product should be user friendly. The customer should be able to use the product easily without the help of expert.

Maintainability:

It refers to the ease with which a product can be maintained in the original condition. Maintainability is measured as mean time to repair (MTTR).

Efficiency:

Efficiency is how much output is taken by different products on giving same input.

Aesthetics:

A product or service should not only perform well but also appear attractive. It also includes color, finish and overall look of product or service.

Serviceability:

It includes:

- How well the customers are treated?
 - How complaints are handled and resolved?
 - How much time you take to resolve problem?
- This feature plays a vital role in service organizations.

TQM Evolution

- **Quality terms (Total Quality Control, Total Quality Improvement, Service Quality Improvement – or titles that relate to their particular area (Quality Maintenance System, Software Quality Assurance**

Quality Control

- **Industrialization (Mass production)**
- **Inspection**
- **W.Shewhart (1924): Statistical Process Control (SPC)**

Quality Assurance

- **George Edwards (inventor of the term)**
- **Assuring quality**
- **Standardization**
- **Policies**
- **Division of responsibility**

Quality Improvement

- Utilizing SPC
- Improving processes
- Reducing waste.
- Individual Improvement projects
- Monitoring variations

Quality Management

- **Improvement Plan at all levels of organization**
- **Internal Motivation**
- **Preventive practice**

Total Quality Management

- All aspects of organization adopting continuous improvement
- Change of organizational culture (shift from internal measures of efficiency towards assessment and improvement of value to the customers)

Strategy for quality evolved with time is given:

Inspection



Quality control

Pre-world war 2

Quality Assurance

Post world war 2



Quality Management

TQM (Evolution of Quality)

In 1985 the Americans came up with the term TQM to represent essentially the Japanese way of quality management.

Quality Control

V's

Quality Assurance

Quality Control

- **Reactive approach**
- **Focuses on product/service**
- **Find defects**
- **Line function**

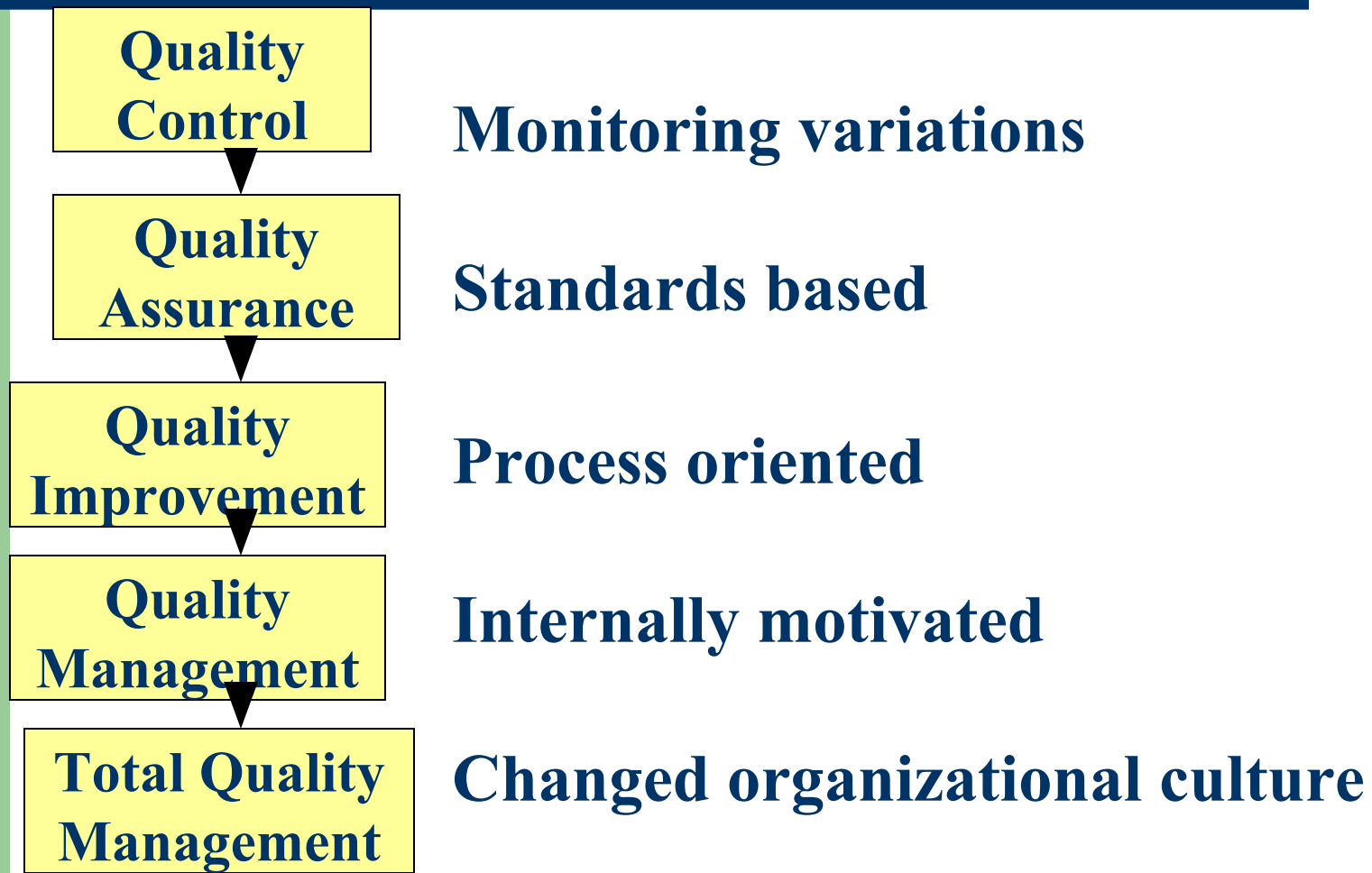
Quality Assurance

- **Proactive approach**
- **Focuses on process**
- **Prevent defects**
- **Staff function**

Quality Management:

QM comprises all activities of the overall management function that determines the quality policy, objectives and responsibilities & implement them by means such as quality planning, quality control, quality assurance and quality improvement with in the quality system.

Stages of TQM



Managing TQM

- **What Does the TQM manager Need to Know?**
 1. **Systems Thinking**
 2. **Understanding Variations**
 3. **Organizational Psychology**

Systems Thinking

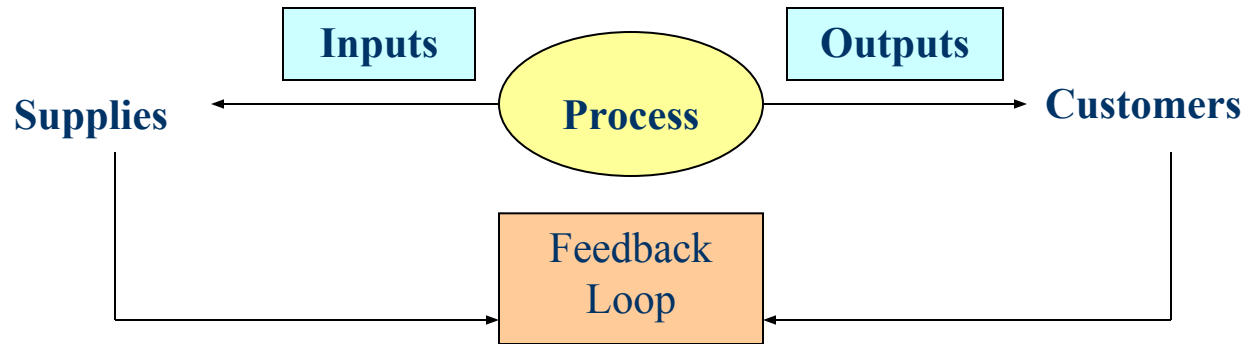


**Effective
Management**

**Understanding
Variations**

**Organizational
Psychology**

Systems Theory



Understanding Variations

- Special cause variation
- Common cause variation
- Interpretation errors
- Process capability
- Example of probability (coin flipping)